Rig day — 2026-10-14 (WEDNESDAY, ACUTE #3): burst-recovered rank, MIMO, hard stop 23:00

> **Built 2026-10-13 from `RIG_DAY_2026-10-08.md` + the 10-12/10-13 prep.** Every command line below is the output of `& $py rig\rig_runcard.py --phase <phase> --day 2026-10-14 …` (lint-checked by `rig\lint_runbook.py`: every script and flag exists). Blanks `<resp> <c1>,<c2> <V> <blk>` are filled by passing `--resp 1,3,4,6 --ctrl 34,40 --baseline <V> --block <dir>` to the runcard, which then prints the lines ready to paste and writes them to `day_2026-10-14\runcard_<phase>.txt`. **What is new since 10-08:** the network path is the USB dongle (verified 10-12: `net_diag` PASS, `find_rz2` ACK, `6_udp_selftest -LiveStim` 8/8; all re-verified 10-13); blocks go **Synapse PC → USB flash drive → this PC** by hand, 2-3 min per block (10-13: the Synapse PC's Ethernet died in the same event as this PC's, it is on Wi-Fi, the lab-NAS route is too slow and a direct SMB share dropped after 20 min; better hardware is an open item) and can be mirrored here with `rig\pull_blocks.ps1 -Source <drive>:\<tank> -Once` (prints `NEW BLOCK:`; the gates read the newest block that actually holds a `.tsq`, so the empty `Desktop\Data\Acute3` folder can no longer be picked); `2_loop` refuses to start the next run after an `AUDIT … FAIL` unless you type `override`; the y-liveness window BEEPS on DEAD; the gap scan can be regenerated on the gate's best pair (`--gap-pair`) and the duty-ID design on the measured operating gap (`--gap-ticks`); references can no longer be overwritten (`--run-offset`/`--force`); `rig\archive_day.ps1` bundles the day; `rig\stim_zero.ps1` after ANY crash; `rig\physio_timer.ps1` keeps the monitoring clock. A full sim rehearsal of this file ran on 10-13 (`rig\rehearse_day.py`, see `LAB_NOTEBOOK_2026-10-13.md` for what it caught).

Staffing: mostly SOLO at the table; the engineer lane is the one-page `HELPER_CARD_2026-10-14.md` anyone can run. **Hard stop on experiments 23:00.** User's call 10-13: **no cuts planned** — the full Tier A → B → C list is the plan; the cut list below is only for a late start. The early/late repeat (6g) is the LAST arm regardless and is never cut; the depth ladder stops at 17:00.

SOLO MODE (decided 10-13 evening): every block transfer is you breaking scrub, so transfer only what the NEXT decision needs and batch the rest.

- **Immediate (the check line reads the block):** every cortical-gate block (Phase G), every thalamic depth block (G′), the quiet block (Phase 1: blacklist + baseline feed every later runcard). Three to seven round-trips all day, 2-3 min each.
- **Batched (the check line reads the capture/log on this PC, the block is for offline analysis):** thwack batteries (galleries), burst probe (its `fit_burst_plant` + `placement_gate` knees come from the capture; the delivery check waits for the batch — the Phase-G′ delivery check already verified the wire), gap, duty-ID, every arm, R sweep, drift. Copy them in two batches: during the Phase-1 settle (gate + quiet blocks are already over; nothing else to do) and at WRAP (burst ≈ 7 GB alone, start it first). The `ingest_block` lines for these phases run at WRAP in block order.
- `run_audit`, `tracking_metrics`, `fit_burst_plant`, `check_dead_time`, `check_mimo_rank`, `analyze_r_sweep` never need the block: run them right after `Summary:` as written.
- The physio timer window sits on this PC's screen facing the table; you answer it yourself.

Conventions: 64 ch (`-InputChannels 64`), Microprobes pair table `--pairs "16:12,15:11,14:10,13:9,4:8,3:7,2:6,1:5"`, recording started LAST, stim enable ON at `go`, **nothing else runs on this PC during a run** (10-13 rehearsal: a 4-min gate run with a data job running alongside lost 11,800 of 26,277 ticks to timeouts — this PC's loop is a 10 ms budget), cpp-primary, uMax 30, blacklist from TODAY's Phase 1 only, y-liveness after every `go`, `run_audit` after every `Summary:`. Python = `C:\Users\brets\Documents\Repositories\PythonIntanAnalysis\.venv\Scripts\python.exe` (set `$py` to it; bare `python` is the Store stub). Every PowerShell needs `Set-ExecutionPolicy -Scope Process Bypass` + `0_preflight` once (MATLAB PATH).

Banked before the day (10-07 … 10-13)

- Designs validated (`DESIGN OK`): `design_rungate.csv` 4.4 min · `design_runburst1.csv` 15.9 min · `design_rungap1.csv` 2.2 min (pair 4 @ 25 µA unless regenerated) · `design_rundutyid_all8.csv` 5.0 min.
- Dongle path verified 10-12 (link 10 Mbps as always; RZ2 10.1.0.100 answers ARP/ping/GET_VERSION; live zero packet delivered). `cpp_controller --selftest` 17/17 (10-13 build); MATLAB bench 28/28 (10-09).
- Dead time = 3 ticks on the 09-10 prep → fitter default `inputDelayTicks 2`; measure again today (`check_dead_time` on the gate probe) and pass onset−1 to the fit.
- Sim rehearsal 10-13 of every phase below (`day_rehearsal1013\rehearsal_rehearsal1013.json`).

08:30 Phase 0 — bring-up, BEFORE induction (20 min)

0a. Pre-induction checklist (tick each; values from the lab's approved protocol where blank)

#	item	PASS	value / note
1	Stimulator battery charged (08-14 false null came from a dead battery) — charged 10-13 evening; confirm the indicator again before induction	☑ 10-13	re-check 08:30: ☐
2	PZ2 amplifier ON and on its charger / battery plan for 14 h (09-10 PZ2-off cost 3 runs)	☐	
3	Synapse stim safety button ON; stim-enable OFF until `go`	☐	
4	Reference + ground wiring checked (cortical array ref, thalamic array ref, animal ground)	☐	
5	Electrode map: cortical grid = row-major, ch1-8 = row 1 (what `placement_gate` assumes) — same array and map as acute #2	☑ 10-13	confirmed by the user 10-13
6	Impedances read in Synapse (note outliers → today's blacklist candidates)	☐	
7	Physiology monitoring per protocol: **every 15 min, heart rate + body temperature on the PhysioSuite** → `physio_timer -EveryMin 15` window open; type the two readings at each beep (e.g. `HR 310 T 37.2`)	☐	PhysioSuite on, probes placed: ☐
8	This PC: no other apps, sleep off (preflight prints it), ≥ 50 GB free (preflight prints it); Defender exclusion for the repo + `Desktop\Data` (admin: `Add-MpPreference -ExclusionPath C:\Users\brets\Documents\Repositories\TDTClosedLoop`) (optional hygiene; the 10-13 rehearsal stall was the rehearsal driver's own undrained output pipe, fixed: three clean 6000-tick sim runs + the rerun show 0 dropped ticks). Never pipe or redirect terminal B's output through another program: run 2_loop.ps1 in its own window	☐	
9	Block transfer is MANUAL by USB flash drive (Synapse PC → drive → `Desktop\Data\Acute_101426\`), 2-3 min per block (10-13 test: 570 MB block; `placement_gate --newest` found it). Drive empty, both PCs' USB ports known. A USB-3 SSD would cut this to ~30 s	☑ 10-13	drive letter here: _____
10	Synapse PC: sleep off, ≥ 50 GB free on the tank drive, Synapse project = the frozen 64-ch circuit with Scle/Plse/sSig/UDP1/Wav1/Wav2 stores	☐	

0b. Network + loop (the runcard `--phase phase0` lines)

```
Set-ExecutionPolicy -Scope Process Bypass; $py = 'C:\Users\brets\Documents\Repositories\PythonIntanAnalysis\.ven
v\Scripts\python.exe'
Get-PnpDevice | Where-Object { $_.InstanceId -match 'VEN_4550' } | Format-List FriendlyName, Status, Problem   #
PO8e must be OK
.\rig\0_preflight.ps1 -InputChannels 64            # PASS: droppedControlTicks=0; also prints sleep setting + fr
ee space
.\rig\net_diag.ps1                                  # auto-detects the dongle ('Ethernet 2'); PASS: the path is
correctly configured
& $py rig\find_rz2.py                              # exactly one device: 10.1.0.100
.\rig\6_udp_selftest.ps1 -RZ2 10.1.0.100 -LiveStim  # 8/8 PASS; E2 live packet is a ZERO -- before the animal is
on the table
```

0c. Block transfer (MANUAL, USB flash drive) + monitoring window

Block transfer is a hand-carried round-trip after every recording that a check line needs (gate blocks, quiet, thwack batteries, burst, arms), **2-3 min per block**: copy the new block folder from the Synapse tank onto the flash drive, walk it over, copy it into `C:\Users\brets\Desktop\Data\Acute_101426\` (the tank folder; blocks are `BSCL20260909BU-261014-<hhmmss>`). Every gate/ingest line uses `--newest`, which picks the newest folder under `Desktop\Data` that holds a `.tsq` (verified 10-13 on the test block `BSCL20260909BU-261013-195629`: 77 s, all stores present, `placement_gate` and `ingest_block` both found it). The helper does the round-trip; a 75 s block is ~570 MB, a 4-min gate block ~1.8 GB, the 16-min burst block ~7 GB (copy it while the next phase runs).

```
# Rig PC, monitoring window (leave open all day):
Start-Process powershell -ArgumentList '-NoExit','-Command',".\rig\physio_timer.ps1 -Day 2026-10-14 -EveryMin 1
5"   # protocol: HR + body temp from the PhysioSuite every 15 min
# Rig PC, second hop (instead of Explorer): mirrors only NEW blocks from the drive and prints NEW BLOCK: <dir>
.\rig\pull_blocks.ps1 -Source <drive>:\<tank> -Once
```

The share path (`pull_blocks -Source \\DESKTOP-G2170R2\SynapseTanks -Tank <subject> -Watch`) stays in the script for a rig with a real link between the two PCs; do not spend surgery time on it.

0d. Card probe (Synapse Preview: frozen 64-ch circuit, PZ2 ON, stim safety ON)

```
.\MpcPo8eUdpClosedLoop.exe 10.1.0.100 . 64 --controller constant --constant-output 0 --skip-udp-send --max-contr
ol-ticks 300
```

PASS = `Streaming. numChannels=64 sampleBytes=4 (float32)`, 300/300, window 6. **Wire gate on a dummy load (optional if time is short — the Phase-G′ probe doubles as it):** `rig_runcard --phase gate --day 2026-10-14 --depth 0` → A/B lines with `-Run gate0`, recording LAST, enable ON at `go` → `run_audit` → `check_impulse_delivery … --design-meta design_rungate_meta.json` → `DELIVERY VERIFIED`, wire == design, 0 missed.

09:30 Induction → surgery (solo lane). Target first recording 15:30; solo-realistic 16:30.

Both craniotomies in ONE drilling sitting; cortical array aimed at the acute-#2 focus (grid rows 5-6, cols 2-7; best channels 34/39/40/47); thalamic target = acute-#2 coordinates, insert to 7.8 mm. Engineer lane idle until Phase G unless a helper is present (then: Synapse Preview, G8 air-thwack 150/150 while you cut, one flash-drive round-trip of the test recording so `placement_gate --newest` prints today's block).

Phase G — cortical placement gate (repeat until GO, ~5-6 min per cycle incl. the 2-3 min drive round-trip)

Seat the planar array → **40 thwacks on D1** (~45 s) → stop recording → flash-drive round-trip (0c) →

```
& $py rig\placement_gate.py --mode cortical --newest
```

GO (z ≥ 8, split-half ≥ 0.80, latency 8-45 ms) → continue. MARGINAL/NO-GO → read the ADJUST line (edge → shift toward it; no focus → move/re-aim), re-thwack 40×, re-run. Log: verdict _____ z _____ focus _____ (cycles: _____). Never the speaker.

Phase G′ — thalamic placement gate (depth ladder, ≤ 3 depths, ~8 min each incl. the drive round-trip; stop the ladder at 17:00)

`& $py rig\rig_runcard.py --phase gate --day 2026-10-14 --depth <d>` prints (d = 1, 2, 3):

```
# A:
.\cpp_controller.exe --mode openloop --play design_rungate.csv --output-count 8 --capture capture_rig_rungate<d
>.csv
# B:
.\rig\2_loop.ps1 -RZ2 10.1.0.100 -InputChannels 64 -TimeoutMs 10 -TickFrames 6 -Run gate<d> -Design design_runga
te.csv -Capture capture_rig_rungate<d>.csv
# after Summary: (recording stopped, NEW BLOCK printed)
& $py rig\run_audit.py --run gate<d> --loop-log loop_rungate<d>.log --capture capture_rig_rungate<d>.csv --desig
n design_rungate.csv
& $py rig\placement_gate.py --mode thalamic --newest --min-amp 13
& $py rig\check_impulse_delivery.py --block <blk> --capture capture_rig_rungate<d>.csv --amps 13 18 25 --pairs
"16:12,15:11,14:10,13:9,4:8,3:7,2:6,1:5" --design-meta design_rungate_meta.json   # wire gate re-bank
& $py rig\check_dead_time.py --capture capture_rig_rungate<d>.csv --y-channel <best ch> --json day_2026-10-14\de
ad_time_gate<d>.json   # -> inputDelayTicks for the fit
& $py rig\ingest_block.py --label "gate depth <d> probe" --day 2026-10-14
```

GO = ≥ 2 RESPONSIVE pairs AND footprint eff rank ≥ 2. Tie-break between depths: more responsive pairs, then higher eff rank, then lower median footprint corr. Depth _____ responsive pairs _____ eff rank _____ best pair _____ best ch _____ dead-time onset _____ (→ `--delay-ticks` = onset−1).

Phase 1 — seal, settle 30 min; quiet capture (60 s) + blacklist + baseline

```
& $py rig\ingest_block.py --label quiet --day 2026-10-14      # runs analyze_quiet_capture -> galleries/<blk>/qu
iet.json
```

blacklist _____ baseline _____ V (→ `--blacklist` and `--baseline` on every later runcard). Eat/drink here.

========== TIER A (minimum publishable day, ~3.5 h) ==========

Phase 2 — thwack battery: 100/site × 6 sites + SHAM (~40 min)

Sites by achievability from the gate focus (acute #2: P1 MP LP P3 strongest; add D1 + one digit). Per block (`rig_runcard --phase thwack --site <SITE> --block <blk>`):

```
& $py rig\ingest_block.py --label "<SITE> thwack" --mode thwack --day 2026-10-14      # SOLO: batched (block at
WRAP)
& $py C:\Users\brets\Documents\Repositories\NNController\scripts\touch\extract_nthw_templates.py --block <blk> -
-site <SITE> --day 2026-10-14 --min-pulse-ms 50 --n-channels 64
```

If first recording slipped past 17:30: 4 sites + SHAM.

Phase 3 — burst-ladder probe (16 min) + gap scan (2 min)

`rig_runcard --phase burst --day 2026-10-14 --blacklist <list> --baseline <V>`:

```
.\cpp_controller.exe --mode openloop --play design_runburst1.csv --output-count 8 --capture capture_rig_runburst
1.csv
.\rig\2_loop.ps1 -RZ2 10.1.0.100 -InputChannels 64 -TimeoutMs 10 -TickFrames 6 -Run burst1 -Design design_runbur
st1.csv -Capture capture_rig_runburst1.csv -Baseline <V>
& $py rig\run_audit.py --run burst1 --loop-log loop_runburst1.log --capture capture_rig_runburst1.csv --design d
esign_runburst1.csv
& $py rig\check_impulse_delivery.py --block <blk> --capture capture_rig_runburst1.csv --amps 6 9 13 18 25 --pair
s "16:12,15:11,14:10,13:9,4:8,3:7,2:6,1:5" --design-meta design_runburst1_meta.json
& $py rig\fit_burst_plant.py --capture capture_rig_runburst1.csv --meta design_runburst1_meta.json --json day_20
26-10-14\burst_plant.json --png day_2026-10-14\burst_plant.png
& $py rig\placement_gate.py --mode thalamic --newest --min-amp 13 --blacklist <list>      # knees per pair
& $py rig\ingest_block.py --label "burst ladder probe" --day 2026-10-14      # SOLO: batched with the delivery c
heck above (block ~7 GB, at WRAP)
```

Gap scan on the gate's BEST pair (`rig_runcard --phase gap --day 2026-10-14 --gap-pair <best pair> --gap-amp 25` regenerates and validates `design_rungap1.csv`; without `--gap-pair` the pre-built pair-4 design plays):

```
.\cpp_controller.exe --mode openloop --play design_rungap1.csv --output-count 8 --capture capture_rig_rungap1.cs
v
.\rig\2_loop.ps1 -RZ2 10.1.0.100 -InputChannels 64 -TimeoutMs 10 -TickFrames 6 -Run gap1 -Design design_rungap1.
csv -Capture capture_rig_rungap1.csv -Baseline <V>
& $py rig\run_audit.py --run gap1 --loop-log loop_rungap1.log --capture capture_rig_rungap1.csv --design design_
rungap1.csv
& $py rig\fit_burst_plant.py --capture capture_rig_rungap1.csv --meta design_rungap1_meta.json --json day_2026-1
0-14\gap_scan.json --png day_2026-10-14\gap_scan.png
```

operating gap = `OPERATING GAP (first ratio >= 0.80)`: _____ ticks (default 24 if unreadable).

Phase 4 — duty-cycled operating-point ID (5 min) → MIMO fit → rank gate

`rig_runcard --phase dutyid --day 2026-10-14 --resp <resp> --gap-ticks <operating gap> --baseline <V>` regenerates `design_rundutyid.csv` + meta on the responsive pairs with the MEASURED gap (omit `--gap-ticks` to subset the pre-built gap-24 design) and must print `DESIGN OK`, then:

```
.\cpp_controller.exe --mode openloop --play design_rundutyid.csv --output-count 8 --capture capture_rig_rundutyi
d.csv
.\rig\2_loop.ps1 -RZ2 10.1.0.100 -InputChannels 64 -TimeoutMs 10 -TickFrames 6 -Run dutyid -Design design_rundut
yid.csv -Capture capture_rig_rundutyid.csv -Baseline <V>
& $py rig\run_audit.py --run dutyid --loop-log loop_rundutyid.log --capture capture_rig_rundutyid.csv --design d
esign_rundutyid.csv --active-u u1,u3,u4,u6
& $py rig\ingest_block.py --label "duty-ID probe" --day 2026-10-14
# fit: rig_runcard --phase fit --day 2026-10-14 --resp <resp> --ctrl <c1>,<c2> --delay-ticks <onset-1>
matlab -batch "cd('C:\Users\brets\Documents\Repositories\TDTClosedLoop'); r = fit_sysid_from_capture('capture_ri
g_rundutyid.csv', struct('useInputs',[<resp>],'useOutputs',[<c1> <c2>],'orders',1:3,'inputDelayTicks',<onset-1
>)); export_plant_lti('plant_duty.lti', r.sys)"
& $py rig\check_dead_time.py --capture capture_rig_rundutyid.csv --u-channels <resp> --y-channel <c1> --lti plan
t_duty.lti   # VERDICT must be OK/centred
& $py rig\check_mimo_rank.py --model plant_duty.lti
.\cpp_controller.exe --mode mpc --model plant_duty.lti --target 0 --check-model   # Observer: ACTIVE, pole < 1,
DC gain printed; exits by itself
```

GO: per-output |corr| > 0.1 on ≥ 2 channels AND rank gate PASS → control channels _____ (channels on DIFFERENT pair footprints — the gate PNG shows them). FAIL → p = 1 on the best channel, pairs = top-2 by knee; the day still yields burst-vs-tonic (6c). `export_plant_lti(path, r.sys)` leaves `AllModels.mat` alone.

Phase 5 — references + tapes + manifest (≤ 10 min)

`rig_runcard --phase refs --day 2026-10-14 --resp <resp> --ctrl <c1>,<c2> --baseline <V>`:

```
& $py rig\build_interleaved_run.py --templates-dir C:\Users\brets\Documents\Repositories\NNController\outputs\Bi
omimeticInversion\touch\Acute_2026-10-14 --sites LP P1 MP P3 D1 SHAM --channels <c1> <c2> --baseline <V> --model
plant_duty.lti --runs 3 --events 100 --total-ticks 22200 --umax 30 --pairs <resp> --out-dir day_2026-10-14
& $py training\closed_loop_sim.py --plant plant_duty.lti --ticks 2000 --pairs <resp> --feature-map <c1>,<c2> --o
utput-count 8 --launch --launch-args --pairs <resp> --feature-map <c1>,<c2> --q-weight 1e12,1e12 --r-weight 100,
100 --control-horizon 20 --umax 30 --reference day_2026-10-14\ref_mix_r1.csv --capture day_2026-10-14\capture_si
m_refcheck.csv
```

→ `wrote 3 run(s)`; sim: `Command varies` and the per-output means move toward the reference.
(`build_interleaved_run` now REFUSES to overwrite an existing `ref_mix_r<N>.csv`: later refs use `--run-offset`.)

Phase 6a — r1 MPC → r1 Choi (paired schedule; ~10 min incl. scoring)

`rig_runcard --phase arm --arm mpc --run 1 --day 2026-10-14 --resp <resp> --ctrl <c1>,<c2> --baseline <V>` (then `--arm choi`):

```
# MPC, terminal A; then B (recording LAST, enable ON at go; the y-liveness window opens and BEEPS on DEAD):
.\cpp_controller.exe --mode mpc --model plant_duty.lti --output-count 8 --pairs <resp> --feature-map <c1>,<c2> -
-q-weight 1e12,1e12 --r-weight 100,100 --control-horizon 20 --umin 0 --umax 30 --reference day_2026-10-14\ref_mi
x_r1.csv --capture day_2026-10-14\capture_mpc_mixr1.csv --max-packets 22200
.\rig\2_loop.ps1 -RZ2 10.1.0.100 -InputChannels 64 -TimeoutMs 10 -TickFrames 6 -Run mpc_mixr1 -Ticks 22200 -Capt
ure day_2026-10-14\capture_mpc_mixr1.csv -Baseline <V>
# Choi:
.\cpp_controller.exe --mode openloop --play day_2026-10-14\design_choi_mix_r1.csv --output-count 8 --capture day
_2026-10-14\capture_choi_mixr1.csv
.\rig\2_loop.ps1 -RZ2 10.1.0.100 -InputChannels 64 -TimeoutMs 10 -TickFrames 6 -Run choi_mixr1 -Ticks 22200 -Cap
ture day_2026-10-14\capture_choi_mixr1.csv -Baseline <V>
# after each Summary:
& $py rig\run_audit.py --run <arm>_mixr1 --loop-log loop_run<arm>_mixr1.log --capture day_2026-10-14\capture_<ar
m>_mixr1.csv --ref day_2026-10-14\ref_mix_r1.csv --active-u u1,u3,u4,u6 --baseline <V>
& $py rig\tracking_metrics.py --capture day_2026-10-14\capture_<arm>_mixr1.csv --ref day_2026-10-14\ref_mix_r1.c
sv --y-channels <c1> <c2> --label <arm>_r1 --json day_2026-10-14\tracking_<arm>_r1.json --png day_2026-10-14\tra
cking_<arm>_r1.png
& $py rig\ingest_block.py --label "<arm> r1 arm" --mode arm --schedule day_2026-10-14\schedule_mix_r1.json --day
2026-10-14      # SOLO: batched (block at WRAP); run_audit + tracking_metrics now
```

`AUDIT <arm>_mixr1: PASS` = fresh@≤100, dropped 0, observer ACTIVE pole<1, offsets=applied, y LIVE, u moves. An MPC arm with `--dist-observer` is a separate, labelled arm (offset-free MPC; not Choi-comparable). A small-ρ arm (`--effort-ratio ≤ 0.01`) should add `--qp-iters 5000`: the 10-13 sim sweep (plant_opfit12_d2, 3000 ticks) converged 91% of ticks at the default cap 200, 99.6% at 1000 and 99.7-100% at 5000, with p99 solve time 0.25 ms and worst 2.3 ms (tick = 10 ms), tracking r0 unchanged (0.668). R=100 arms keep the default.

DECISION POINT A (clock ~19:30, or whenever Tier A is banked): tired → jump to WRAP. Else Tier B.

========== TIER B (+1.5 h): 6b → 6c → 6d ==========

6b — decoupled-target MIMO run (only if the rank gate PASSED)

`& $py rig\build_decoupled_ref.py --ref day_2026-10-14\ref_mix_r1.csv --out day_2026-10-14\ref_decoupled_r1.csv` → MPC as 6a with `--reference …ref_decoupled_r1.csv --capture … capture_mpc_decoupled.csv` (`-Run mpc_decoupled`); score per the sidecar `_schedule.json` (active channel tracks, inactive holds baseline = selection shown).

6c — burst-vs-tonic paired arms (schedule r1, two MPC runs back to back)

Run A = 6a's MPC (`--umin 0`: the controller may drop the carrier to zero between events = sparse/burst regime); run B = same model/refs with `--umin <knee uA>` (carrier never below the recruitment knee = tonic regime): `rig_runcard --phase arm --arm mpc --run 1 --suffix _umin --umin <knee>`. Report r0, charge ABOVE HOLD, and the Phase-7 drift probe for both.

6d — r2 MPC → r2 Choi (`--run 2`).

DECISION POINT B (~21:00): tired → WRAP. Else Tier C.

========== TIER C (+1 h): 6e → 6f ==========

6e — interleaved R sweep: ONE 13-min run, R ∈ {100, 30, 10} in 20-event blocks

Build a 120-event reference as run **r4** (never overwrite r1 — 6g needs it):

```
& $py rig\build_interleaved_run.py --templates-dir C:\Users\brets\Documents\Repositories\NNController\outputs\BiomimeticInversion\touch\Acute_2026-10-14 --sites LP P1 MP P3 D1 SHAM --channels <c1> <c2> --baseline <V> --model plant_duty.lti --runs 1 --events 120 --total-ticks 26600 --umax 30 --pairs <resp> --out-dir day_2026-10-14 --run-offset 3
& $py rig\rig_runcard.py --phase rsweep --run 4 --day 2026-10-14 --resp <resp> --ctrl <c1>,<c2> --baseline <V>
# prints build_r_schedule, A (with --r-schedule), B, run_audit, analyze_r_sweep
```

`R SCHEDULE OK`; after the run every row of the `analyze_r_sweep` table says `rW0 ok`.

6f — r3 MPC → r3 Choi (`--run 3`). NN clone arm not prepared; skip.

========== WRAP (starts no later than 22:15, every tier) ==========

6g — EARLY/LATE repeat of r1, both arms, identical tapes/refs (~8 min). Never cut.

`rig_runcard --phase arm --arm mpc --run 1 --suffix _late …` and `--arm choi … --suffix _late` (labels `mpc_mixr1_late` / `choi_mixr1_late`; the originals are never overwritten).

Phase 7 — drift re-probe (duty-ID replay, 2 min) → stim zero → archive

`rig_runcard --phase drift …`:

```
.\cpp_controller.exe --mode openloop --play design_rundutyid.csv --output-count 8 --capture capture_rig_rundutyid_late.csv --max-packets 12000
.\rig\2_loop.ps1 -RZ2 10.1.0.100 -InputChannels 64 -TimeoutMs 10 -TickFrames 6 -Run dutyid_late -Ticks 12000 -Capture capture_rig_rundutyid_late.csv -Baseline <V>
& $py rig\run_audit.py --run dutyid_late --loop-log loop_rundutyid_late.log --capture capture_rig_rundutyid_late.csv --expect-ticks 12000
& $py rig\check_dead_time.py --capture capture_rig_rundutyid_late.csv --u-channels <resp> --y-channel <c1> --lti plant_duty.lti   # gain/lag drift vs the fit: ______
.\rig\stim_zero.ps1                                  # 8 words x 2 s of zeros
.\rig\archive_day.ps1 -Day 2026-10-14                # zip of captures/logs/ledger/galleries -> Desktop\Data\Archive
```

23:00 hard stop. Ledger rows are auto-appended by `ingest_block` into `BLOCK_LEDGER_2026-10-14.md`. Tomorrow: copy the archive zip + `Desktop\Data\Acute_101426` to the external drive / lab NAS (the Synapse PC keeps the originals); commit + push both repos.

Cut list (one list; cut from the bottom)

Tier A (6a + 6g) is never cut. If first recording > 17:30: battery 4 sites + SHAM, Tier A only + 6g. If > 18:30: skip the gap scan (use gap 24) and run 6a + 6g only. Otherwise cut in this order: 6f (r3) → 6e (R sweep) → 6d (r2) → 6c (burst-vs-tonic) → 6b (decoupled).

Red flags (say them immediately)

- The liveness window BEEPS / says `DEAD` (or `run_audit` says `y=DEAD`) → amplifier/PZ2 off. Stop the run.
- `AUDIT …: FAIL` → the next `2_loop` refuses to start until you type `override`; the surgeon decides.
- **Any crash or abort of a stim run** → **`.\rig\stim_zero.ps1` first**, then diagnose.
- B never prints `policy=fresh` by tick ~100 → Ctrl+C in B, restart B, `go` again.
- `Skipping … to position` in B → Synapse recording was started too early; stop, restart B, start recording LAST.
- A gate/ingest line prints `newest block:` with a time that is not the recording just stopped → the copy has not landed (recording still open, or copied before Synapse closed it); it graded a STALE block.
- Never re-run Synapse Detect mid-day (08-29); never start another program on this PC during a run.

Endurance rules

Water/food during the 16-min burst probe and the 30-min settle; sit for all arm runs; a helper runs the HELPER_CARD lane whenever present; the physio timer window is the clock for the animal.